CSIS DAYS 2026

Beyond the Building: A Probabilistic Framework for GPS Assignment Under Positional Uncertainty

Hafiyyan Hilmy Fawwaz · Makoto Chikaraishi
Hiroshima University

full presentation

1

MOTIVATION

where did they *really* stop?

In a city, GPS error is often bigger than the distance between a building, the street in front of it and the open space next to it.

The usual fix is to pick the nearest building. But urban researchers have long shown that daily life also happens outside buildings.

> **Jacobs** *the street*
> The Death and Life of Great American Cities, 1961
>
> **Gehl** *the space between*
> Cities for People, 2010
>
> **Whyte** *the small plaza*
> The Social Life of Small Urban Spaces, 2010

THE GAP

one answer, *too sure* of itself

Picking one building also hides how unsure the GPS is. For people who are **moving**, map matching already gives probabilities. For people who **stay**, nobody had done this. And nobody had covered all three types: building, street and open space.

OUR THREE QUESTIONS

RQ1 How can we build GPS error into a method that gives probabilities?

RQ2 How often are its answers the same as the survey answers?

RQ3 Does a more complex model help, or does it fail on new stays?

2

HOW WE DID IT

let the uncertainty *speak*

1 Find candidates
Buildings, street pieces and open spaces within 10 m of the stay.

2 Give four scores
Each candidate gets the scores below.

3 Add them up
Each score has a weight. One number per candidate.

4 Make probabilities
A logit model gives one probability per place type.

Time near — share of the stay spent close to it
Distance — how far it is from the stay center
Longest stretch — longest unbroken time at it
Direction — movement along the street

The weights are learned from the survey answers. The final model keeps three of these scores and adds two more inputs.

THE DATA

Building
Z-Map TOWN building outlines

Street
MAPPLE street center lines, made as wide as the street

Open space
All the ground that is left

FIELD SURVEY IN HIROSHIMA

People visited planned stops and wrote down the place type, and when they arrived and left. Each person carried three or more phones, so we have more GPS data per stop.

We downsampled the survey GPS points to match MILES, the GPS data we want to apply the model to: 152,602 points became 15,687.

11 people, 2 sessions each
3–5 phones per person
280 real stops
907 stays with a known answer

building 531 | street 185 | open space 191

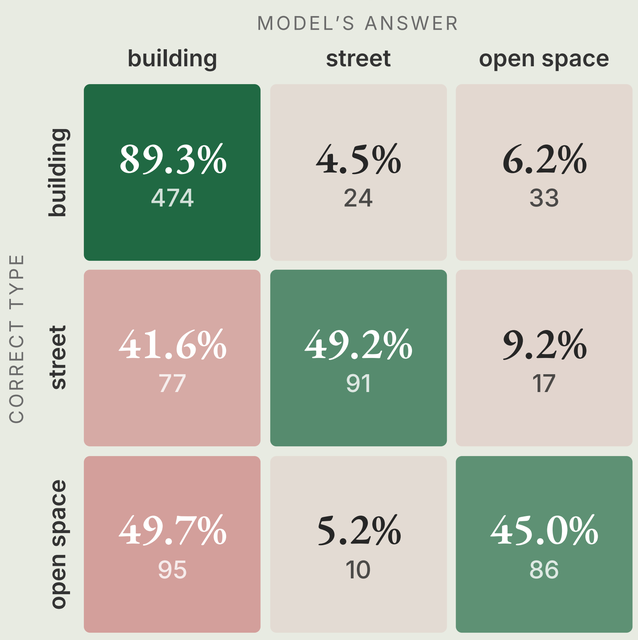
3

WHAT WE FOUND

right *7 times in 10*, even on new stays

71.8% of new stays get the right place type. "New" means the model did not learn from them.

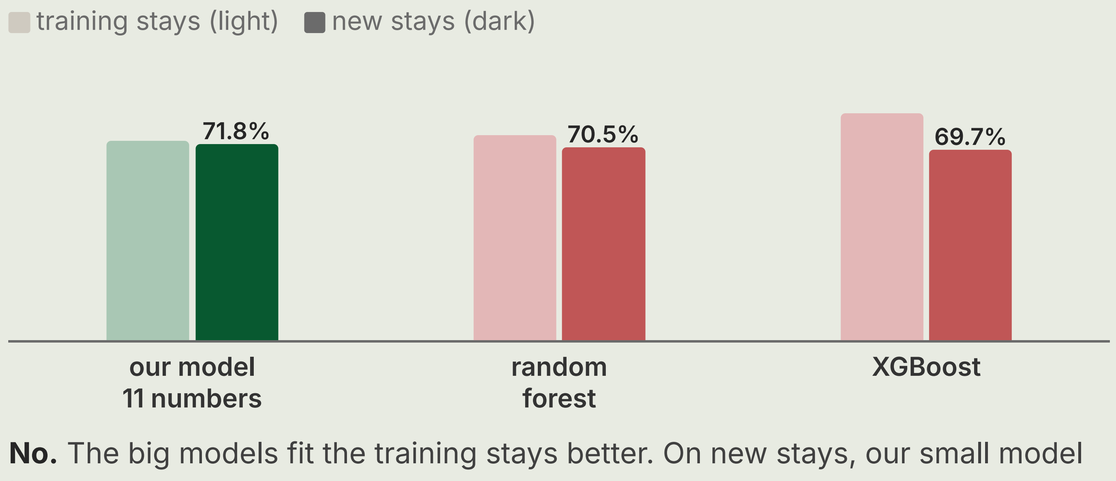
WHAT IT GETS RIGHT, AND WRONG



2×
Buildings are found about **twice** as often as streets or open spaces.

THE TEST

is a bigger model *better?*



No. The big models fit the training stays better. On new stays, our small model is the best.

WHY ARE STREETS MISSED?

The map cuts one street into many short pieces. A street stay has many street pieces nearby, and the model counts this against the street.

13.1 street pieces near a street stay
6.0 same type near a building stay
4.2 same type near an open-space stay

New. A probability for building, street and open space, for people who stay in one place.

Simple is enough. 11 numbers beat two machine-learning models on new stays.

Fixable. For streets, the weak point is how the map cuts them. It is not the GPS data.

REFERENCES

1. Jacobs, J. (1961). *The Death and Life of Great American Cities*.
2. Gehl, J. (2010). *Cities for People*. Island Press.
3. Whyte, W. H. (2010). *The Social Life of Small Urban Spaces*.
4. Quddus, M. A., et al. (2007). Current map-matching algorithms for transport applications. *Transp. Res. C*, 15.
5. Bierlaire, M., et al. (2013). A probabilistic map matching method for smartphone GPS data. *Transp. Res. C*, 26.
6. Kato, R., et al. (2024). Evaluating and validating stay point detection algorithms. *Procedia Comput. Sci.*, 238.
7. Wu, H., et al. (2026). Object-oriented spatial statistics for human activity space estimation. *arXiv:2605.08509*.
8. McFadden, D. (1973). Conditional logit analysis of qualitative choice behavior.

hafifawwaz@hiroshima-u.ac.jp